NANDHAKUMAR G

Mechanical Engineer Undergrad

Mechanical Engineering undergraduate specializing in the integration of advanced CAD modeling, autonomous systems, and additive manufacturing. Proven ability to optimize mechanical performance through data-driven analysis and machine learning.

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IMPACT SNAPSHOT & KEY MILESTONES

8	3	4
CERTIFICATIONS	INDUSTRY INTERNSHIPS	ENGINEERING PROJECTS

EXECUTIVE FOCUS

- ✓ **Multidisciplinary Engineering:** Specializing in the strategic integration of advanced CAD modeling, autonomous robotic systems, and additive manufacturing.
- ✓ **Data-Driven Optimization:** Bridging the gap between digital simulation and physical prototyping with data-driven analysis and machine learning to optimize mechanical design performance.
- ✓ **Industrial Execution:** Committed to driving engineering innovation through a collaborative, sustainable approach to complex, high-reliability system design.

CORE COMPETENCIES

MECHANICAL DESIGN

Digital Prototyping | Model Optimization | Parametric Modeling | Additive Manufacturing

SYSTEMS ENGINEERING

Autonomous Robotics | Sensor Integration | Finite Element Analysis | Control Systems

TOOLS & METHODS

Fusion 360	SolidWorks	Siemens NX	AutoCAD
Python	C++	MATLAB	COMSOL

PROFESSIONAL ATTRIBUTES

Problem Solving:
Developed and implemented engineering solutions for complex structural and mechanical problems, validating layouts with mathematical modeling.

Team Collaboration:
Worked effectively in diverse engineering teams for project success, interfacing seamlessly with technicians and senior project engineers.

Time Management:
Managed multiple engineering projects simultaneously while adhering to strict schedules and prototype deadlines.

PROFESSIONAL EXPERIENCE (INTERNSHIPS)

Industrial Engineering Trainee
Steel Authority of India Ltd. (SAIL), Salem Steel Plant — December 2025

- Gained industrial exposure to integrated steel manufacturing processes: raw material handling, continuous casting, hot rolling mills, and finishing.
- Observed and documented the operational mechanics of heavy metallurgical machinery, automated material flow systems, and industrial quality control.
- Learned maintenance practices, safety protocols, and operational compliance standards followed in core manufacturing plants.
- Developed practical knowledge of shop-floor workflows, industrial safety execution, and public-sector engineering workflows.

Product Design Intern
Femlogic Technologies Pvt. Ltd., Chennai — June 2025 - July 2025

- Contributed to the structural assembly design of the Vande Bharat train seating modules, improving components interfaces and complying with DFM standards.
- Acquired hands-on exposure to structural product design, 3D parametric CAD modeling, and design validation methodologies.
- Collaborated closely with senior mechanical engineers, developing expertise in industry-standard design validation.

Production Engineering Intern
K.M. Knitwear Pvt. Ltd., Tirupur — July 2024 - August 2024

- Analyzed existing shop-floor production lines, proposing bottlenecks optimizations that drove a 10% boost in operational efficiency.
- Co-designed and refined structural packaging configurations for corrugated box manufacturing, enhancing shipping protection.
- Assisted in routine maintenance routines and mechanical troubleshooting of industrial machinery, boosting overall equipment reliability by 15%.

SELECTED PROJECT PORTFOLIO

Project EcoFloat | Niral Hackathon

LEAD DEVELOPER & DESIGNER | ROS · IoT · Fusion 360

- Spearheaded development of a dual-function autonomous robotic boat engineered to simultaneously remove physical aquatic trash and detect chemical runoff.
- Implemented a real-time toxicity sensing framework powered by IoT and ROS, designed to transition marine cleanup from reactive manual labor to a proactive, data-driven automated system.
- Architected the platform's physical design in Fusion 360, prioritizing solar-powered endurance and scalability for deployment by municipal and port authorities.

Regenerative Braking System for Electric Vehicle

LEAD RESEARCH & DEVELOPER | MATLAB Simulink · BMS

- Designed and developed a regenerative braking system to improve energy efficiency in an electric vehicle prototype.
- Converted kinetic energy during braking into electrical energy, optimizing recovery using MATLAB Simulink models.
- Integrated motor-generators and battery management systems (BMS), ensuring smooth operation and increasing battery lifespan by 20%.

Advanced Aquatic Monitoring System

SYSTEMS ENGINEER | Hyperspectral Imaging · Sensor Fusion

- Designed and developed an autonomous floating platform equipped with a synthesized high-resolution hyperspectrometer to conduct real-time environmental analysis of water bodies.
- Processed hyperspectral imaging data to accurately detect and quantify microplastic concentrations, while simultaneously mapping chlorophyll levels to assess aquatic health and potential algal blooms.
- Integrated advanced sensor data algorithms, improving ecological monitoring efficiency and providing precise, actionable data for marine conservation and water quality management.

Hemispherical CPV Solar Tracking System

MECHANICAL DESIGN ENGINEER | COMSOL Multiphysics · Solar Vector Math

- Developed a hemispherical CPV solar tracking system with dual-axis tracking and adaptive weather control to improve energy efficiency.
- Designed hemispherical shell-shaped panels to enhance light absorption and angular coverage, validating the design through COMSOL Multiphysics® simulations.
- Optimized solar tracking calculations, making the system ideal for compact and wearable energy harvesting applications.

EDUCATION QUALIFICATIONS

Bachelor of Mechanical Engineering — 2023 - Present
St. Joseph's College of Engineering, Chennai, Tamil Nadu
Core Coursework: Thermodynamics, Fluid Mechanics, Machine Design, Control Systems

Higher Secondary Education (Class XII) — 2011 - 2023
Shri Swamy Matriculation of Higher Secondary School, Salem, Tamil Nadu
Computer Science Stream with Distinction

PROFESSIONAL CREDENTIALS

- **Autodesk Certified Fusion Designer** — Professional CAD Design Credential
- **Certified Solidworks Designer** — Mechanical Modeling & Validation Spec
- **Certified Siemens NX CAD Designer** — Advanced Parametric Modeling Specialist
- **AutoCAD for 2D Modelling** — Technical Drafting & Detailing Standard
- **Certified in Python Advanced** — Computational Data Structures & Integration
- **Certified Frontend Developer** — Core Interface Architect (HTML/CSS/JS)
- **Blender for 3D Modelling** — Complex Surface Mesh Optimization
- **C++ Programming** — Object-Oriented Software & Logic Design

LANGUAGES

- **English** — Fluent
- **Tamil** — Native